

1000 MW(e) SODIUM COOLED FAST BREEDER REACTOR STUDIES

C. A. PURSEL* and L. E. LINK*

1. Introduction

Design studies are an essential part of the development of an economically competitive fast breeder reactor (LMFBR). The studies evaluate and crystallize the supporting research and development effort, identify problem areas, and explore the complex relationships among safety objectives, the demand for reliability of operation, and low power costs. Conversely, changes in safety criteria, reliability requirements, and projections of nuclear power plant economics in the decades ahead, all affect current designs.

The ultimate goal of the LMFBR programme is the development of safe, reliable, and commercially acceptable nuclear power plants. The progress to date towards this goal can be divided into four categories: (a) proof of principle—EBR-I;⁽¹⁾ (b) demonstration of feasibility—EBR-II⁽²⁾ and Fermi;⁽³⁾ (c) exploration of technical limits—the first four design studies; and (d) power-cost-oriented design studies.

Historically, the first successful demonstration of the breeding concept was the NaK cooled, metallic fuelled EBR-I, which was built and operated by the Argonne National Laboratory (ANL) under the sponsorship of the U.S. Atomic Energy Commission (USAEC). The EBR-I was followed by the EBR-II, which is now demonstrating the technical feasibility of a total system complex consisting of an LMFBR power plant coupled with on-site fuel reprocessing and fabrication, and the Fermi reactor, which is a part of an investor owned utility system. The EBR-II and Fermi are also metallic fuelled reactors. By 1962, projections of long range commercial demand had shifted to sizes up to 1000 MW(e) and sufficient work had been done to support the possibility that a ceramic fuelled liquid-metal cooled fast breeder reactor with long fuel residence times might also be technically feasible and economically competitive, even with off-site reprocessing and fuel fabrication. As a consequence, in 1963 a series of studies was initiated to explore

* Argonne National Laboratory, Illinois 60440.

the technical feasibility and economic potential of 1000 MW(e), ceramic fuelled LMFBR power plants.

2. The Four Design Studies of a 1000 MW(e) Ceramic Fuelled LMFBR

Four design studies of a 1000 MW(e) ceramic fuelled LMFBR were published in January 1964.⁽⁴⁻⁷⁾ An evaluation of these four design studies was issued in December 1964.⁽⁸⁾ These reports have been widely distributed and discussed in papers and addresses, at both national and international conferences during the past 2 years.

We will restrict our discussion of the four studies to those portions that resulted in follow-on studies and changes in the supporting research and development programme.

The major conclusions of the four design studies were that: (1) with further studies and development, it appeared to be technically feasible to design and operate an economically competitive ceramic fuelled LMFBR large enough to produce 2500 MW(th), and (2) there appeared to be several practical ways to design a ceramic fuelled core so that even severe nuclear excursions could be self-limiting.

The four design studies were limited to the core, reactor vessel, and their appurtenances. It was recommended that the next phase of the design studies should be directed towards developing a total plant concept to establish the research and development programmes required for prototype designs. An early phase of the design study would involve analyses to: (1) examine the effect of core outlet temperature on plant cost, (2) select fuel handling and transfer methods for reliability, short transfer time, and low cost, (3) evaluate a single tank primary system arrangement versus a system having the primary sodium pumps and intermediate heat exchangers outside the reactor tanks, (4) decide between conventional containment and a hot-cell-type structure, and (5) define the areas that have the greatest potential for reducing plant cost.

Because of technical uncertainties, no choice could be made between mixed oxides and mixed carbides as a fuel for future ceramic fuelled fast reactors; however, all else being equal, the carbide fuels would have a higher breeding ratio than the oxides (1.40 compared to 1.20) and fuel-cycle costs lower by some 0.2 mill/kWh (assuming government ownership of fuel). By the time the studies had been completed, legislation had been passed which required private ownership of reactor fuels by 1971, a time when fast breeder reactors would become of interest. It was recommended that this change should be reflected in future fuel-cycle cost calculations, and that the effect on future core designs should be assessed.

The variation in core designs, in the four studies, was primarily a direct consequence of nuclear safety considerations, particularly the sodium void

effect and the Doppler coefficient. The establishment of realistic safety criteria was recommended, with particular emphasis on sodium voiding and the dynamics of coupled cores.

At the outset of the studies there were reservations about the technical and economic justification for the ground rule requiring a steam temperature of 1000°F (538°C). Since these reservations still remained after completion of the studies, a cost-oriented study of preferred temperatures was recommended.

3. Follow-on Research and Development

The results of the four 1000 MW(e) design studies have significantly influenced LMFBR research and development.

Sodium Pump Test Facility

The total mass flow rate of sodium through the core of a 1000 MW(e) reactor is near 10^8 lb/h (7.6×10^5 kg/min). Because sodium pumps of the required sizes have not been developed, a sodium pump test facility has been authorized which will test pumps up to one-third of the total mass flow rate. Head requirements, operating transients, and sodium temperatures will simulate those anticipated in the 1000 MW(e) plants.

Large Valves for Sodium Service

Because the 1000 MW(e) plants and the sodium pump test facility will require much larger valves than are now available, a programme has been initiated to develop large valves for sodium service. Typical parameters are: 30 in. (76 cm) diameter, 100 lb/in² (7.8 atm) design pressure, and 1200°F (649°C) design temperature. Butterfly, swing-gate check, and combination shut-off and flow control valves are being considered for development.

Analysis of Sodium Pipe Installations

Existing analytical techniques are not adequate to predict the strains, stresses, and fatigue life of large thin wall pipe installations at temperatures approaching 1000–1200°F (538–649°C). Preliminary analyses of rigid (expansion loop) and non-rigid (bellows) piping systems have already been completed. The dimensions, operating pressure, operating temperature, and pipe materials were taken from the 1000 MW(e) design studies of Allis-Chalmers⁽⁴⁾ and Combustion Engineering.⁽⁵⁾

Controlled-expansion Fuel Assembly

The Westinghouse design study⁽⁷⁾ describes a controlled-expansion (CEX) fuel assembly that consists of a combination of metallic fuel pins and unrestrained carbide fuel pins. Such an assembly would provide a predictable

axial thermal expansion in response to power transients in ceramic fuelled fast reactor cores. Westinghouse is studying the transient response of the CEX fuel assembly.

Basic Mechanisms in Sodium

A programme to develop data for basic mechanisms in sodium service is now in progress. The objective is to determine the allowable loadings, corrosion, friction factors, clearances, and operating characteristics of basic mechanisms used in control-rod drives, fuel handling equipment, and fuel hold-down devices in typical sodium service. Operating conditions were selected from the design studies.

4. Follow-on Design Studies of 1000 MW(e) LMFBR Power Plants

During the past 2 years, General Electric⁽⁹⁾ (GE) and Westinghouse⁽¹⁰⁾ (W) have updated and supplemented their original studies. New studies of 1000 MW(e) LMFBR plants have been completed by Atomics International⁽¹¹⁾ (AI) and ANL.⁽¹²⁾ United Engineers and Constructors (UE&C) has completed studies of the capital and operating costs for the electrical portions of 1000 MW(e) plants for a selected range of steam cycles.⁽¹³⁾ This work was used as input to the AI and ANL studies.

The study by AI: (1) explored the safety, feasibility, and economics of a range of coupled or modular core concepts; (2) evaluated a core configuration which could use either oxide or carbide fuel; (3) made a preliminary, but significant, power-cost-oriented analysis of the thermodynamics of steam and sodium cycles, and (4) estimated the fuel-cycle and power costs for a reference LMFBR. The overall geometry and arrangement of the reference modular core is shown in Fig. 1. Some of the more interesting aspects of the core and its arrangement will be discussed in later sections.

The Argonne National Laboratory study is an application of the integrated reactor and fuel reprocessing and fabrication concept now being demonstrated by the EBR-II. The reference core (Fig. 1) consists of six modules. The central zone of each module is surrounded by blanket regions that are edged inside and outside by beryllium and a reflector of stainless steel. The fuel in the central, active region is an alloy of uranium, plutonium, and titanium. The fuel in the blanket is metallic uranium. The tubular cladding of both fuels is an alloy of vanadium and titanium.

A later GE study⁽⁹⁾ reports a refinement of their original design concept. The new design (Fig. 2) is based on updated safety and design criteria and on the results of analyses of the behaviour of the plant in accident situations. The study also considers the effects of uncertainties in physics data and fuel performance on power costs.

A low core height to diameter (H/D) ratio of 0.172 was selected for the original design to simplify the mechanical design and the analyses of core

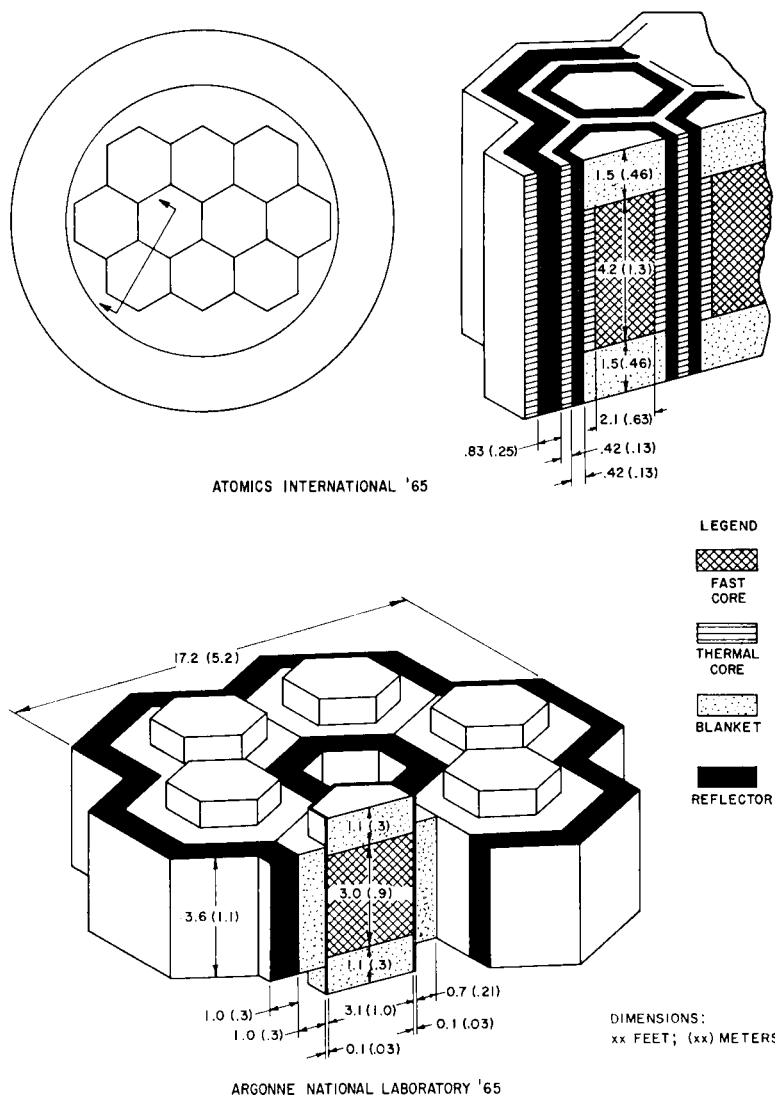


FIG. 1. Reactor core arrangements.

transients. The revised design purposely enhances the inherent core safety characteristics at the expense of economics. The H/D ratio is reduced to 0.116 to further increase neutron leakage. The mean fuel volume fraction

has been increased from 25.4% to 38.6% to increase the Doppler effect. The original design used 7.5 vol.% BeO, uniformly distributed throughout the core to soften the neutron spectrum and increase the Doppler effect.

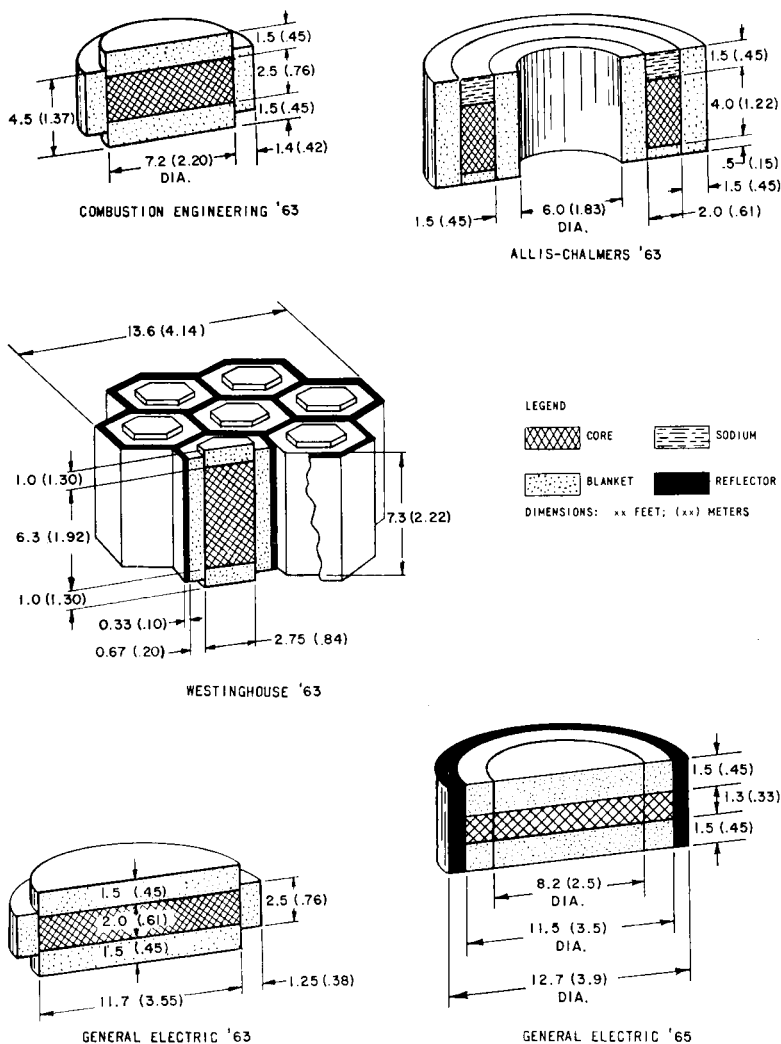


FIG. 2. Reactor core arrangements.

The later design uses 10.5 vol.% in a central zone and 3.5 vol.% in an outer zone to increase the radial power flattening and the relative Doppler effect in the central region. The resulting calculated net local power coefficient is negative everywhere in the core.

TABLE 1. DESIGN AND PERFORMANCE CHARACTERISTICS OF 1000 MW(e) CORES*

Reactor study reference	A-C ('63) (4)	CE ('63) (5)	GE ('63) (6)	W ('63) (7)	GE ('65) (9)	W ('65) (10)	ANL ('65) (12)	AI ('65) (11)
Core geometry	Annulus	Flat cyl.	Flat cyl.	7 module	Flat cyl.	7 module	6 module	10 module
Volume, litres	7720	2900	6030	7050	3930	7050	4680	2800 ^b
Fuel, vol. %	29.4 ^c	25.6 ^c	25.4	29.4 ^c	38.6	29.4 ^c	33	38.8 ^c
Sodium, vol. %	40.0	66.5	46.4	55.1	36.2	55.1	40	47.0
Structure, vol. %	30.6	7.9	18.8	15.5	17.5	15.5	24	14.2
Moderator, vol. %	—	—	7.5	—	7.1	—	—	—
Fuel								
Type (core and ax. bl.)	Oxide	Carbide	Oxide	Carbide	Oxide	Carbide	Metal	Oxide
Type (blanket)	Oxide	Carbide	Oxide	Oxide	Oxide	Oxide	Metal	Oxide
Rod o.d., mm	7.6	7.6	6.3	7.6	6.8	7.6	5.3	6.3
Clad material	316	19-9DL	316	316L	316	316L	V-20Ti	304
Density (smear), %	91	85	81	81	88	81	71	85
Enrichment (avg.), % Pu ^d	20.8	13.0	18.0	15.7	17.9	15.7	15.5	25 ^b
Temperature, °C								
Na inlet	510	454	427	525	427	454	382	399
Na outlet	649	602	593	649	593	593	538	580
Fuel (max)	2540	1427	2600	1195	2720	1332	781	—
Clad (max)	721	760	721	760	—	719	608	696
Coolant velocity (max), m/sec	3.1	6.1	3.4	8.2	5.5	9.1	7.3	5.6 (avg.)
Heat flux, w/cm ²								
Average	11.7	38.4	17.0	16.4	19.2	21.4	12.9	18.9
Maximum	18.0	58.9	35.9	39.7	37.2	51.7	21.8	26.2
Avg. power density, kW/litre	282	695	365	308	—	390	450	495 ^b
Avg. spec. power, kW/kg U+Pu	114	209	170	91	170	111	120	105
Lineal power, W/cm								
Average	282	925	345	394	407	512	216	364
Maximum	430	1430	745	955	795	1240	328	745
Breeding ratio	1.20 ^e	1.41 ^e	1.21 ^e	1.38 ^e	1.18	—	1.48	1.31

^aCoupled fast thermal. ^bFast region only. ^cNo allowance for control. ^dDoes not include Pu-240 or Pu-242.^eNormalized values from ref. (5).

* All studies 2500 MW(th) except W ('65) which is 3440 MW(th).

The Westinghouse studies:⁽¹⁰⁾ (1) develop an updated core design based on private ownership of fuel and a lower temperature of core outlet sodium, and (2) present analyses of the transient response of the coupled modular core concept for a range of postulated accidents. The geometry of the newer design (Fig. 2) is almost identical to that of the original design. The thermal performance is upgraded by 37% (from 2500 MW(th) to 3440 MW(th)).

Selected design and performance and characteristics of the eight 1000 MW(e) core designs are shown in Table 1.

5. Fuel-cycle and Power Costs

A summary of the estimated fuel-cycle and power costs from the eight design studies is shown in Table 2. The original studies assumed a 4 $\frac{3}{4}$ % use

TABLE 2. FUEL-CYCLE AND POWER COSTS (mills/kWh)

	1963 studies 4 $\frac{3}{4}$ % use charge				1965 studies 10% Pu inventory charge			
	A-C ⁽⁴⁾	CE ⁽⁵⁾	GE ⁽⁶⁾	W ⁽⁷⁾	GE ⁽⁹⁾	W ⁽¹⁰⁾	AI ⁽¹¹⁾	
Fabrication	0.470	0.261	0.344	0.309	0.421	0.308	Oxide 0.284	Carbide* 0.344
Interest on fuel fabrication	0.050	0.106	0.029	0.010	0.125	0.027	0.017	0.020
Use or inventory charge	0.300	0.155	0.191	0.318	0.431	0.572	0.539	0.584
Shipping and reprocessing	0.280	0.191	0.194	0.188	0.20	0.188	0.181	0.192
Pu credit	(0.320)	(0.412)	(0.188)	(0.550)	(0.175)	(0.526)	(0.353)	+(0.515)
Total fuel cycle	0.78	0.30	0.57	0.28	1.002	0.569	0.668	0.625
Capital cost (14% cap., 80% plant factor)					2.5	—	2.40	2.40
Operating costs					0.34	—	0.266†	0.266
Total power costs					3.84	—	3.334	2.291

* Incremental costs. Cores have equal metal densities.

† Includes 0.037 and 0.029 mills/kWh for working capital and graphite replacement respectively.

charge on the value of plutonium. The later studies estimate an annual charge of 10%. The later studies by GE and W^(9, 10) also incorporate more refined physics calculations and safety analyses.

Neither the fuel-cycle costs nor the power costs have been normalized or evaluated and both are therefore of questionable use in cross-comparisons.

However, they do reflect the best estimates of the individual firms. The differences between the estimates are as likely to be the result of differences in conservatism of the estimator as they are to be differences inherent in the designs.

The absolute value of any of the fuel-cycle cost estimates is quite uncertain. For example, each of the estimates assume 10 at. % burnup. Limited irradiation data tend to support burnups of this magnitude, but there is an absence of supporting data from irradiation of fuel assemblies in operating reactors. The absolute value of reprocessing costs is particularly uncertain since the costs are based on extrapolations of USAEC reference charges for a hypothetical aqueous processing plant that is designed for low enriched uranium from thermal reactor fuels and not on high-plutonium high burnup fast reactor fuels. Comparative cost studies being made by ANL indicate that a coupled pyrochemical processing plant has about half the cost of an aqueous processing plant of equal capacity. The process is applicable to ceramic as well as to metallic fuels.

6. Fuel Irradiations

The three uranium-plutonium fuels that have had significant development are the oxides, carbides, and metals. The oxides are the first choice, mainly because of the successful experience with UO_2 in thermal reactors. The carbides are also of interest because of their higher density and thermal conductivity. Recent irradiation experience⁽¹⁴⁾ has demonstrated that metallic fuels, with reasonable fuel element designs, can be considered seriously for burnups to 10 at. %. Important characteristics of the fuel designs used in the studies are shown in Table 1. The fuel-rod diameters fall in the range 0.2–0.3 in. (0.51–0.76 cm) with the median above the average of the range. The smear density of the fuel inside the cladding is between 80% and 90% (except near 70% for metal fuels). The enrichments, considering Pu-239 and Pu-241, are all near 15–20%. If Pu-240 and Pu-242 were also considered, the enrichments would be some 5% higher. The total Pu concentration in the AI design⁽¹¹⁾ is probably near 30%. All studies use one of the stainless steels for cladding wherever it is compatible with the fuel to be contained. The maximum lineal power outputs are 13–24 kW/ft (426–787 W/cm) for the oxides, 29–44 kW/ft (951–1440 W/cm) for the carbides, and 10 kW/ft (328 W/cm) for metal fuel. All fuels are assumed to produce 10 at. % burnup per cycle.

The influence of the 1000 MW(e) design studies on the planning of irradiation programmes is evident in Tables I, II, and III of ref. (15), the proposed Core-I irradiations for the FARET reactor. The dimensions of the fuel pins and fuel assemblies, the fuel and cladding materials, and the operating conditions all closely match those in the design studies. It was proposed to extend

the fuel irradiations to include vibratory compacted fuels as well as cladding materials stronger than the stainless steels for some of the sub-assemblies.

The FARET project is no longer active. The USAEC now proposes to build a fast fuels test facility (FFTF).⁽¹⁶⁾ The major differences in the operating conditions of the two reactors is that the FFTF would have a neutron flux of 10^{16} , whereas the FARET flux would have been lower by a factor of 2 or 3. The FARET fuels would have required enrichment with U-235 if the plutonium concentrations were to have been characteristic of those projected for large reactors, while the FFTF can operate without further U-235 enrichment of the test fuels. The FFTF concept uses an enriched driver core, with the test fuel in loop facilities within the core; in FARET, the core loading would have been entirely of fuel under test.

Since the purpose of the FFTF is to test fuels over the range of interest for future large LMFBR nuclear power plants, the design studies were used as guides in formulating its functional requirements. Performance requirements extracted from the studies include: $0.3\text{--}1.0 \times 10^{16}$ neutron flux, 500–2800 W/g of Pu fuel power density, 800–1400°F (426–760°C) core sodium temperatures, 2–6 ft (61–183 cm) fuel assembly length, and both sealed and vented fuel pins.

As of January 1, 1966, a specific fuel irradiation programme had not yet been established for the FFTF, but it is reasonable to assume that the programme will be strongly influenced by the 1000 MW(e) design studies.

7. Accident Analyses and Reactor Safety

Upon completion of the first four studies further study of core safety characteristics was recommended. Specifically mentioned were studies of the magnitude of the variations in reactivity with sodium removal, the manner in which it was removed, the dynamics of coupled cores, and the minimum amount of moderator required from safety considerations.

The results of work on these topics were presented at the International Conference on Safety, Fuels, and Core Design in Large Fast Power Reactors held at Argonne National Laboratory, October 11–14, 1965. The work reported by AI,⁽¹¹⁾ GE,⁽⁹⁾ and W^(10, 11) is summarized in the following sections.

Coupled Core Dynamics

Atomics International⁽¹¹⁾ and W^(10, 17) each studied the dynamics of coupled cores. Both used the kinetics model proposed by R. Avery.⁽¹⁸⁾

The leakage of neutrons into each module from the adjacent module, that is the fraction by which the module would be subcritical in the steady state if the cores were not coupled, is called the coupling reactivity. Coupling

reactivity is least at the beginning of core life (with depleted fuel in the blankets), but increases with the build-up of plutonium in the blankets.

Westinghouse used a two-core module to gain insight into the effects of neutron cross-coupling. Reactivity was always inserted into one module, called the driver module; the other module was designated the passive module. A coupling reactivity of 0.4% was used for beginning-of-life modules and an upper limit of 4% for end-of-life modules. Four per cent is about twice the coupling that would exist in the W reference core with Pu in the blankets. A range of ramp, sinusoidal, and parabolic reactivity insertions into the driver module was considered. The response of an isolated, or fully

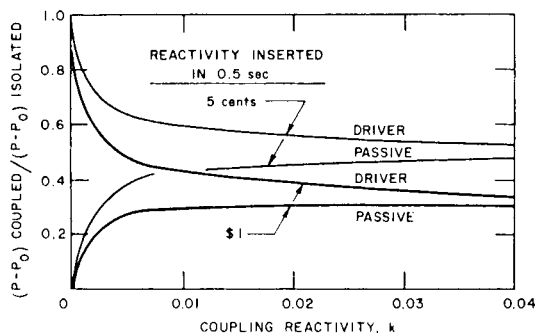


FIG. 3. Power rise ratio at 1.2 sec.

decoupled, module was obtained for each postulated reactivity insertion to provide a basis for assessing the effects of reactivity sharing between the modules. Thus, for the steady state initial conditions, the decoupled reactivity of the cores was 1.0 for the reference case, 0.996 for depleted uranium in the blankets, and 0.96 with twice the coupling estimated for blankets containing 3 at. % Pu. In all cases the only negative reactivity feed-back considered was the Doppler effect, $T dk/dT$ of -0.00383 . Two values of the neutron coupling lifetime were assumed: 0.35×10^{-6} sec and 0.35×10^{-5} sec.

The gross results of the W study are shown in Fig. 3. The following conclusions were reached: (1) for all cases considered, the neutron coupling lifetime has a negligible effect on the power split between the driver and the passive core, (2) a small amount of coupling reactivity, for example 0.1% has a pronounced effect in reducing the power rise from the insertion of reactivity into one of the modules, and (3) the effectiveness of any given coupling reactivity increases with the magnitude of the reactivity inserted.

Westinghouse also concluded that their reference design of seven modules may be overly conservative. A smaller number of larger modules would result in higher internal breeding ratios, lower inventory requirements, shorter fuel doubling times, and lower fuel-cycle costs.

Atomics International, after studying a variety of cores with coupled arrangements of fast fuel, blanket fuel, and moderator, chose a modular split blanket flux-trap core (MOFT) for their reference design (Fig. 1). The MOFT core has ten modules. Each module consists of a central, fast region surrounded by a blanket region that is split by a row of graphite moderator elements. The blanket pins have an average equilibrium concentration of $3\frac{1}{2}\%$ Pu. To determine the effect of variations of cross-coupling, calculations were performed using steady state reactivity values of 0.9, 0.97,

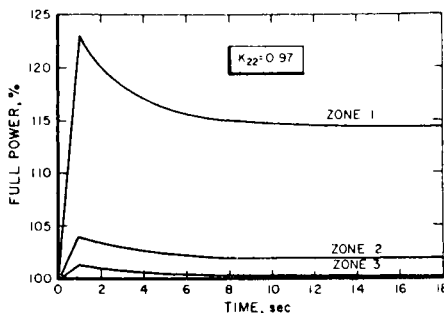


FIG. 4. Power response—\$1 inserted in Zone 1 in 1 sec.

and 0.99. The Doppler coefficients, $T dk/dT$, used in the calculation were: -0.0023 in the fast region, -0.00902 in the inner blanket, -0.00678 in the outer blanket, and -0.018 for the total core. The calculated sodium void effect was -236ϕ with the fast section voided, -245ϕ with the total core voided, and $+53\phi$ for the most reactive distribution of voids. The calculated prompt-neutron lifetime is 2.0×10^{-6} sec.

More than twenty accidents were analysed. Some were time-dependent responses to postulated combinations of malfunctions of equipment and misoperation. Others were responses to arbitrarily selected reactivity insertions. The latter class of accidents was used to give insight into the response of coupled modules.

Figure 4 illustrates the power response of the three zones with a reactivity of \$1 in 1 sec, with no action of the control system, and with coolant flow rate and inlet temperatures held constant. Figure 5 illustrates the power response under the same conditions except for a reactivity coupling of 0.10.

It was concluded by AI that the modular core concept has inherently better safety response characteristics to all accidents involving insertion of reactivity into one module. Examples of such accidents are: (a) the ejection of any single control rod, (b) flow blockage in a single channel, and (c) dropping a fuel element into a slightly subcritical core.

The MOFT reference design may be overly conservative. The conservatism

could be reduced by adding one ring of fuel assemblies to the fast section of each module. Fuel-cycle costs would be reduced by about 0.2 mills/kWh, the inventory required to reach criticality would be reduced by approximately 25%, and the breeding ratio would be increased slightly. The Doppler coefficient would be somewhat more negative but the maximum sodium void reactivity would be increased from 53¢ to \$2.

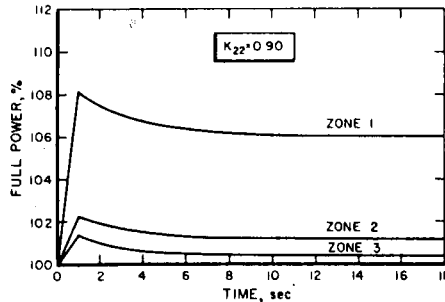


FIG. 5. Power response—\$1 inserted in Zone 1 in 1 sec.

Core Compaction Accidents

Core compaction accidents were studied by AI⁽¹¹⁾, GE⁽⁹⁾, and W⁽¹⁰⁾. Studies of such accidents are conducted to set an upper limit on the design features that are provided to cope with the release of explosive energy from the core. In performing the analyses, combinations of events are assumed that will lead to core melting. The top portion is then assumed to fold over on the bottom portion to form a critical mass, which is dispersed by the force of the resulting explosion.

Since accidents of this nature require that the control system does not function, they usually are described as hypothetical. Nevertheless, such accidents are studied, and an estimate is made of the maximum amount of energy that will be available to do work in units of MW/sec.

Atomics International assumed the following series of events: (1) the loss of coolant flow to the entire core with simultaneous failure of the protective system, (2) voiding of the fast regions, and (3) fuel meltdown after a few minutes. Previous analyses had shown that the modules would not all reach the melting point simultaneously. It was assumed, for the purpose of the analysis, that the two central modules participated in the meltdown, that the compacted fuel became a homogeneous sphere in each of the two modules, and that the resulting explosion eliminated the possibility of meltdown accidents in the other modules.

The total energy release was calculated to be 360 MW/sec, and it was

assumed that half of this (180 MW/sec) became available for mechanical work.

General Electric assumed that flow blockage of the central fuel channel was followed by local sodium voiding and fuel compaction in the channel. They also assumed, as an independent event, the loss of all coolant flow to the reactor, followed by sodium voiding in all of the channels, melting of

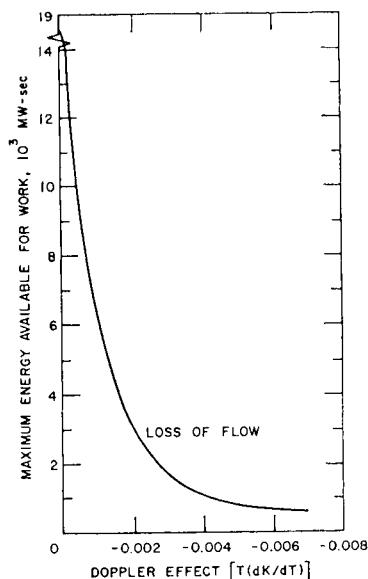


FIG. 6. Available energy from core compaction.

the fuel progressively in a radial direction from the centre of the core, compaction of the molten fuel to 100% of its theoretical density, followed by release of energy and dispersion of the fuel. The calculated amount of energy available to do work is shown as a function of the Doppler coefficient in Fig. 6. The energy available to do work is 1030 and 647 MW/sec, assuming isentropic values of $-T dk/dT$ of 0.004 and 0.006 respectively.

Westinghouse assumed that if, for some unidentified reason, fuel melting did occur, and if the fuel slumped uniformly throughout the core within the fuel cladding, thereby increasing its density from 90% to 95%, the reactivity addition in an isolated module would be \$18. Folding over the upper half of the core into the bottom half, including slumping within the core, assuming free fall, would add \$75 worth of reactivity in 0.3 sec. The total calculated energy release for the isolated module would be 2100 MW/sec.

Accidents with Scram

Each of the three studies contain preliminary analyses of a wide range of possible accidents that were usually classed as improbable but not hypothetical in that the plant protective, or scram, system was assumed to be operable. None of the analyses led to the release of energy from a second criticality.

The most damaging accident was the instantaneous blockage of coolant flow to one channel since no instrumentation exists that would allow preventive action before the fuel reached the melting point. However, in the GE core,⁽⁹⁾ with its softened spectrum and low H/D ratio, the net reactivity increase from sodium voiding and complete meltdown of a single fuel assembly is less than 40¢.

8. Plant Thermodynamics—Sodium and Steam Conditions

The evaluation report⁽⁸⁾ suggested that: (1) the low fuel-cycle costs projected for the large ceramic fuelled reactors might permit lower cost plant equipment than would be required if the high temperature, associated with high thermodynamic efficiency, were necessary in order to reduce fuel cycle costs; and (2) both the maximum fuel cladding temperature and the turbine throttle temperature should be established by a cost oriented, optimization study which balances costs associated with reduced material properties with increasing temperature, the effect on fuel-cycle costs, and the variation in the capital costs of the plant with a reduction in plant cycle efficiency.

Although no study of the magnitude implied by these recommendations has been completed, aspects of the problem have been studied and reported.

United Engineers and Constructors have completed a study of the total cost for the engineering and installation of the steam-electric portion of five 1000 MW(e) and five 500 MW(e) nuclear plants.⁽¹³⁾ Results of the 1000 MW(e) estimates are summarized in Table 3. The costs do not include items such as the cost of land, switchgear, and interest on money during construction.

The ANL design⁽¹²⁾ is based on the results of a cost optimization of the nuclear power plant. Costs of power-generation islands were taken from the UE&C report. Fuel-cycle costs of 0.8 mills/kWh were used in the economic analyses. The lowest power costs resulted from a combination of a core outlet sodium temperature of 1000°F (538°C), steam pressure and temperature of 1800 lb/in² (123.5 atm) and 900°F (482°C), and steam reheat to 900°F (482°C).

The optimization by AI⁽¹¹⁾ is also partially based on the results of the UE&C study. Direct quotations for turbine generator sets were received from GE. The study assumes fuel-cycle costs of approximately 0.5–1.0

mills/kWh. The lowest power costs resulted from a combination of a core outlet sodium temperature of 1050°F (566°C), steam pressure and temperature of 2400 lb/in² (164.3 atm) and 900°F (482°C), and steam reheat to 900°F (482°C). AI recommends further studies to determine if total power costs would be further reduced with fewer feedwater heaters, lower core outlet temperatures, and a temperature rise through the core of less than 350°F (194.4°C).

TABLE 3. COST OF 1000 MW(e) STEAM-ELECTRIC ISLANDS

Turbine inlet pressure	Turbine inlet temperature	Reheat temp.	Net MW(e)	Total cost ($\times 10^{-6}$)	Cost per net kW(e)
3500 lb/in ² (239.1 atm)	1000°F (538°C)	1000°F (538°C)	1027.9	\$ 47.5	\$ 46.26
2400 lb/in ² (164.3 atm)	1000°F (538°C)	1000°F (538°C)	1027.2	46.2	44.98
1800 lb/in ² (123.5 atm)	900°F (482°C)	900°F (482°C)	1029.5	49.8	48.37
1000 lb/in ² (69.0 atm)	546.4°F (Sat) (285.8°C)	—	1052.4	63.8	60.62

The Westinghouse study is based on a core outlet sodium temperature of 1100°F (649°C), core temperature rise of 250°F (138.8°C), and steam temperatures between 950°F and 1000°F (510°C and 538°C). No steam cycle optimization studies are reported.

The General Electric study is based on a core sodium outlet temperature of 1100°F (649°C), and a core temperature rise of 300°F (166.7°C).

9. Summary

Conceptual designs of 1000 MW(e) LMFBR power plants are going through an evolutionary process.

The original designs were based on the achievement of selected technical limits, such as high thermodynamic efficiency, high core power density, or high breeding ratio within the constraints of the safety requirements. Later designs tend to concentrate on safety requirements or low power costs. The system parameters that are now evolving appear to be more nearly within currently known technical limits.

Greater emphasis is being placed on reducing the reactivity gains from sodium voiding and increasing the Doppler effects. Loosely coupled modular

cores are used in more conceptual designs, core height-to-diameter ratios are being decreased and the use of some moderating material in the reactor is gaining acceptance. There is a general belief that each of these design provisions may reflect an overemphasis on specific accidents that may ultimately be shown to be hypothetical.

Core sodium outlet and steam temperatures are decreasing with a consequent decrease in power costs and an increase in both the effectiveness of the Doppler coefficient and in the fuel cladding strength. Although earlier unreported design studies were based on temperature rises as high as 500°F (278°C), the trend is for the temperature rise through the core to decrease. The increased cost for the pumps and the pumping power required for the lower temperature rises through the core appears to be more than offset by the decrease in heat exchanger area.

Of the ceramic fuels, the oxides continue to be preferred over the carbides. Several designs have provision for the later substitution of carbide fuel. Metallic fuels are no longer so severely limited in burnup capabilities by swelling characteristics; there now is concrete evidence that reasonable metallic fuel element designs are capable of 10 at. % burnup.

References

1. H. V. LICHTENBERGER, F. W. THALGOTT, W. Y. KATO and M. NOVICK, Operating experience and experimental results obtained from a NaK-cooled fast reactor, *Proceedings of the International Conference on the Peaceful Uses of Atomic Energy, Geneva, 1955*, Vol. 3, p. 345, United Nations, New York (1956).
2. L. J. KOCH, H. O. MONSON, W. R. SIMMONS, M. LEVENSON, F. VERBER, E. HUTTER, R. A. JAROSS, T. R. SPALDING, J. R. SIMANTON, and A. LOVOFF, Construction design of EBR-II: An integrated unmoderated nuclear power plant, *Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy, Geneva, 1958*, Vol. 9, p. 323, United Nations, New York (1958).
3. A. AMOROSI and J. G. YEVICK, An appraisal of the Enrico Fermi reactor, *Proceedings of the International Conference on the Peaceful Uses of Atomic Energy, Geneva, 1958*, Vol. 9, p. 323, United Nations, New York (1958).
4. Allis-Chalmers Mfg. Co., Atomic Energy Division, Milwaukee Atomic Power Development Associates, Inc., Detroit, and Babcock & Wilcox Company, Atomic Energy Division, Lynchburg, Virginia, *Large Fast Reactor Design Study*, USAEC Report ACNP-64503 (Jan. 1964).
5. Combustion Engineering, Inc., Nuclear Division, *Liquid Metal Fast Breeder Reactor Design Study*, USAEC Report CEND-200, Vols. I and II (Jan. 1964).
6. General Electric Company, *Liquid Metal Fast Breeder Reactor Design Study (1000 MW(e) UO₂-PuO₂ Fueled Plant)*, Vols. I and II, USAEC Report GEAP-4418 (Jan. 1964).
7. R. B. STECK (Comp.) *Liquid Metal Fast Breeder Reactor Design Study, Final Report*, USAEC Report WCAP-3251-1, Westinghouse Electric Corp. (Jan. 1964).
8. Reactor Division of Chicago Operations Office, *An Evaluation of Four Design Studies of a 1000 MW(e) Ceramic Fueled Fast Breeder Reactor*, USAEC Report COO-279 (Dec. 1964).
9. K. P. COHEN and G. L. O'NEILL, Safety and economic characteristics of a 1000 MW(e) fast sodium-cooled reactor design, *Conference on Safety, Fuels, and Core Design in Large Fast Power Reactors, October 11-14, 1965*, sponsored by Argonne National Laboratory and United States Atomic Energy Commission, Argonne, Illinois.

10. J. A. WRIGHT *et al.*, Conceptual design and preliminary accident analysis of a sodium cooled, carbide fueled, large modular fast reactor, *Conference on Safety, Fuels, and Core Design in Large Fast Power Reactors, October 11-14 1965*, sponsored by the Argonne National Laboratory and United States Atomic Energy Commission, Argonne, Illinois.
11. D. T. EGGEN *et al.*, *Feasibility Study of a 1000 MW(e) Sodium Cooled Fast Reactor*, USAEC Report NAA-SR-11378, Atomics International (June 1965).
12. L. E. LINK *et al.*, A large metal-fueled fast reactor study, *Conference on Safety, Fuels, and Core Design in Large Fast Power Reactors, October 11-14, 1965*, sponsored by Argonne National Laboratory and United States Atomic Energy Commission, Argonne, Illinois.
13. United Engineers and Constructors Inc., *Engineering Study for Large Steam-Electric Generating Units Applied to Argonne Fast Breeder Reactor Studies*, Vol. 1, and Sub-contract No. 31-109-38-1717 (Nov. 1964).
14. R. E. MACHERY *et al.*, Metal fuels for fast reactors, *Conference on Safety, Fuels, and Core Design in Large Fast Power Reactors, October 11-14, 1965*, sponsored by Argonne National Laboratory and United States Atomic Energy Commission, Argonne, Illinois.
15. P. J. PERSIANI *et al.*, *FARET Core I Fuel Irradiation Program and Reference Design*, USAEC Report ANL-7106, Argonne National Laboratory (Oct. 1965).
16. E. R. ASTLEY *et al.*, The fast-flux test facility design concept, *Trans. Am. Nucl. Soc.* **8** (2), 607 (Nov. 1965).
17. W. E. GUNSON, F. M. HUB, and R. S. DALEAS, Transient response of coupled fast reactor cores, *Conference on Safety, Fuels, and Core Design in Large Fast Power Reactors, October 11-14, 1965*, sponsored by Argonne National Laboratory and United States Atomic Energy Commission, Argonne, Illinois.
18. R. AVERY, Theory of coupled reactors, *Proceedings of the Second United Nations International Conference on the Peaceful Uses of Atomic Energy, Geneva, 1958*, Vol. 12, p. 182, United Nations, New York (1958).